

A Packing of 17 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 2.941950. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.931715, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.941950 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 17$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.931715, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 17/16 = 2.942308$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N17.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

294195093827062395355696659381941204668560538838627685142262238172685568
531720842195286348234690319856283132337411249059546857538846569671054849
8314076209147836156338262958546674740159417107911

denominator:

999999999999110261209134097160517505999518208498034126080772038941038940
457323281332952273797816340006387929455587289968244885099418453446182545
870317074937144899423621201327642954276764155745

B Coordinates

17 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N17.txt`.

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